



Create AI

DATA MINING PROJECT

Hotel Booking Cancellation Prediction: Identifying Key Risk Factors and Building Predictive Models to Reduce Revenue Loss in the Hospitality Industry

Course: Data Mining

Program: Bachelor of Science in Business Administration

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Motivation

The hospitality industry operates on thin margins, and booking cancellations represent one of its most persistent operational challenges. When a reservation is cancelled, the hotel loses potential revenue, disrupts staffing plans, and faces uncertainty in room allocation. In recent years, the rise of online travel agencies and flexible booking platforms has made cancellations even more frequent and harder to anticipate. Data mining offers a concrete solution: by identifying the patterns that precede a cancellation, hotels can take proactive steps before it happens. This project is motivated by the belief that predictive analytics, applied correctly to real booking data, can turn an unmanageable operational risk into an actionable business decision.

Problem Background

Hotel booking cancellations have increased significantly with the rise of online travel platforms, where customers can book multiple hotels simultaneously and cancel for free. This creates a major problem for hotels: rooms go unsold at the last minute, revenue forecasts become unreliable, and operational planning suffers. The core issue is that hotels have no systematic way to distinguish high-risk bookings from reliable ones at the time of reservation. Data mining addresses this directly by learning patterns from historical booking data and assigning a cancellation risk to each new reservation.

Related Work

The dataset used in this project was introduced by Antonio, de Almeida, and Nunes (2019), who also tested several machine learning models on it. They found that Random Forest achieved the best performance with AUC above 0.90 and identified lead time, deposit type, and market segment as the strongest predictors these are results that the present project confirms independently.

Sánchez-Medina and C-Sánchez (2020) worked on the same dataset and argued that interpretability matters as much as accuracy: hotel managers need to know which variables to act on, not just whether a booking is risky. Their work reinforces the importance of feature importance analysis.

Höpken et al. (2021) showed that gradient boosting methods such as LightGBM can match or exceed Random Forest performance on hospitality data, which motivates including multiple models in the comparison rather than defaulting to a single approach.

Across all these studies, three findings are consistent: ensemble methods outperform linear models, lead time and deposit type are the most reliable predictors, and model value depends on how clearly results translate into actionable hotel policy.

1. Dataset Description

This project uses the Hotel Booking Demand Dataset, originally published by Antonio, de Almeida, and Nunes (2019) and made publicly available via Kaggle. It contains 119,390 booking records across two hotel types ; a city hotel and a resort hotel ; both located in Portugal. Each row represents one booking and includes 32 variables covering reservation timing, customer profile, booking channel, room assignment, pricing, and deposit policy. The target variable is **is_canceled**, a binary indicator taking the value 1 if the booking was cancelled and 0 otherwise.

2. Problem Statement

The project addresses a binary classification problem: can we predict, at the time of booking, whether a reservation will be cancelled before arrival? This question matters because early identification of high-risk bookings allows hotels to intervene ; through targeted deposit rules, reminders, or pricing adjustments ; before revenue is lost.

The main research question is: **which factors most strongly drive hotel booking cancellations, and which predictive model identifies them most reliably?**

3. Data Preparation

Data quality directly determines model performance, so preparation was treated as a core phase rather than a preliminary step.

The first priority was eliminating target leakage. The variables **reservation_status** and **reservation_status_date** were removed at the outset, as they directly encode the final booking outcome and would give any model an unrealistic advantage.

The dataset was then cleaned by removing exact duplicates and addressing domain-specific inconsistencies. Missing values in *agent* and *company* were filled with 0 to represent the absence of an intermediary, and observations with zero total guests or negative ADR were discarded as invalid records.

Categorical variables were processed according to cardinality. Features with low cardinality were encoded using OneHotEncoder to avoid introducing artificial ordering, while high-cardinality variables were encoded using OrdinalEncoder to limit dimensionality expansion. Numerical variables were scaled using StandardScaler, and missing values were imputed using

the median. All transformations were implemented within a ColumnTransformer pipeline fitted exclusively on the training set to prevent data leakage.

A correlation-based feature selection step was applied to numerical variables, removing those with negligible linear association with the target ($|r| < 0.02$) in order to reduce noise for linear models.

Finally, the dataset was split into training and test sets using an 80/20 stratified split, ensuring that both sets preserved the original class distribution.

Handling Class Imbalance: The class imbalance in this dataset is moderate at 27.3% cancellations versus 72.7% non-cancellations. Rather than applying resampling techniques such as SMOTE, which generate synthetic observations and are best reserved for severe imbalance, the strategy adopted here relied on three complementary measures: (1) stratified train/test splitting, ensuring the class ratio is preserved in both training and test sets; (2) stratified cross-validation in GridSearchCV, so every fold maintains the same distribution; and (3) AUC-ROC as the primary evaluation metric, which is robust to class imbalance because it evaluates model discrimination across all possible thresholds rather than at a single cut-off. Threshold tuning was subsequently applied to the best-performing model to further improve recall on the minority class without resampling.

4. Methods and Analysis

Four modeling approaches were considered within an end-to-end pipeline framework that included all preprocessing steps.

Logistic Regression served as the interpretable baseline model. It estimates cancellation probability as a linear function of the predictors and benefits from feature scaling applied during preprocessing.

Random Forest was the primary candidate. Hotel cancellation behaviour is unlikely to follow simple linear relationships; for example, the effect of lead time may interact with deposit type and market segment. Random Forest captures these complex, non-linear patterns through an ensemble of decision trees.

Decision Tree: was included for interpretability. A tuned single tree produces explicit decision rules that are easy to communicate to managers, although it is more prone to overfitting than the ensemble model.

Principal Component Analysis (PCA): was explored within the pipeline to assess whether compressing the feature space could improve the performance of a linear model. PCA was fitted exclusively on the training data and applied as a preprocessing step before Logistic Regression. However, it was ultimately retained as a diagnostic tool rather than a final modelling approach, as it did not improve predictive performance.

All models were tuned using GridSearchCV with 5-fold Stratified Cross-Validation, optimising for AUC-ROC throughout.”

5. Results and Interpretation

5.1 Exploratory Data Analysis

Target Distribution

The dataset exhibits a moderate class imbalance, with 50,570 non-cancelled bookings (72.7%) and 19,000 cancelled ones (27.3%) in the training set. This distribution confirms that cancellations are less frequent but still represent a significant operational issue for hotels. Because of this imbalance, accuracy alone would be misleading, as a model could achieve high accuracy by simply predicting the majority class. To properly evaluate performance, AUC-ROC and F1-score were used as the primary metrics, as they better capture the model’s ability to identify cancellations.

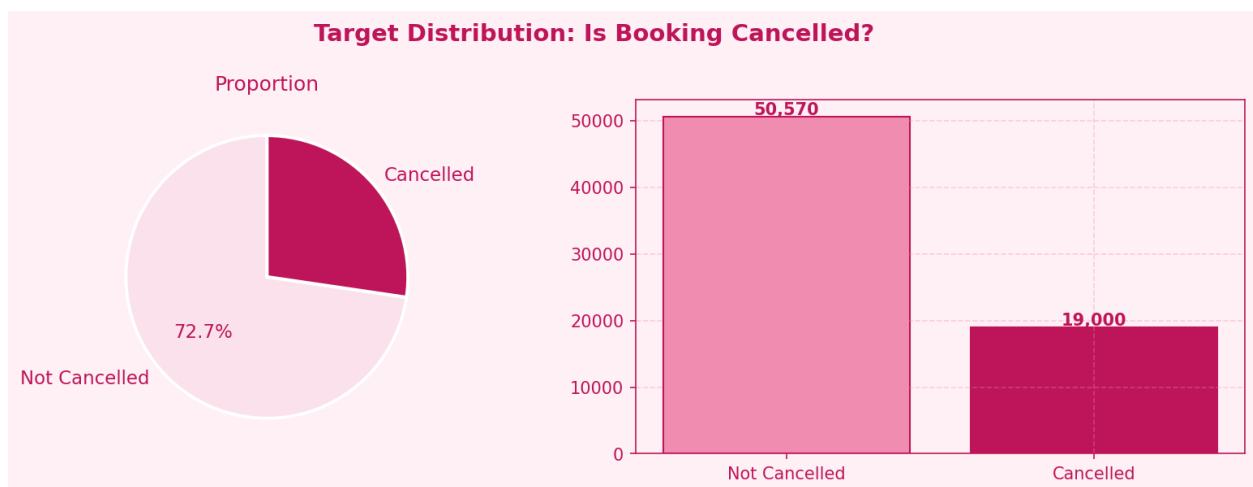


Figure 1 “Target Distribution”

Correlation Analysis

The correlation analysis shows that **lead_time** is the strongest positive predictor of cancellation ($r \approx +0.18$), meaning bookings made further in advance are more likely to cancel. **ADR** and **arrival_date_year** also show weaker positive relationships.

On the other hand, **required_car_parking_spaces**, **total_of_special_requests**, and **is_repeated_guest** are negatively correlated, indicating that more committed customers cancel less.

Overall, correlations are weak ($|r| < 0.2$), suggesting that cancellations are driven by **combined effects rather than a single variable**.

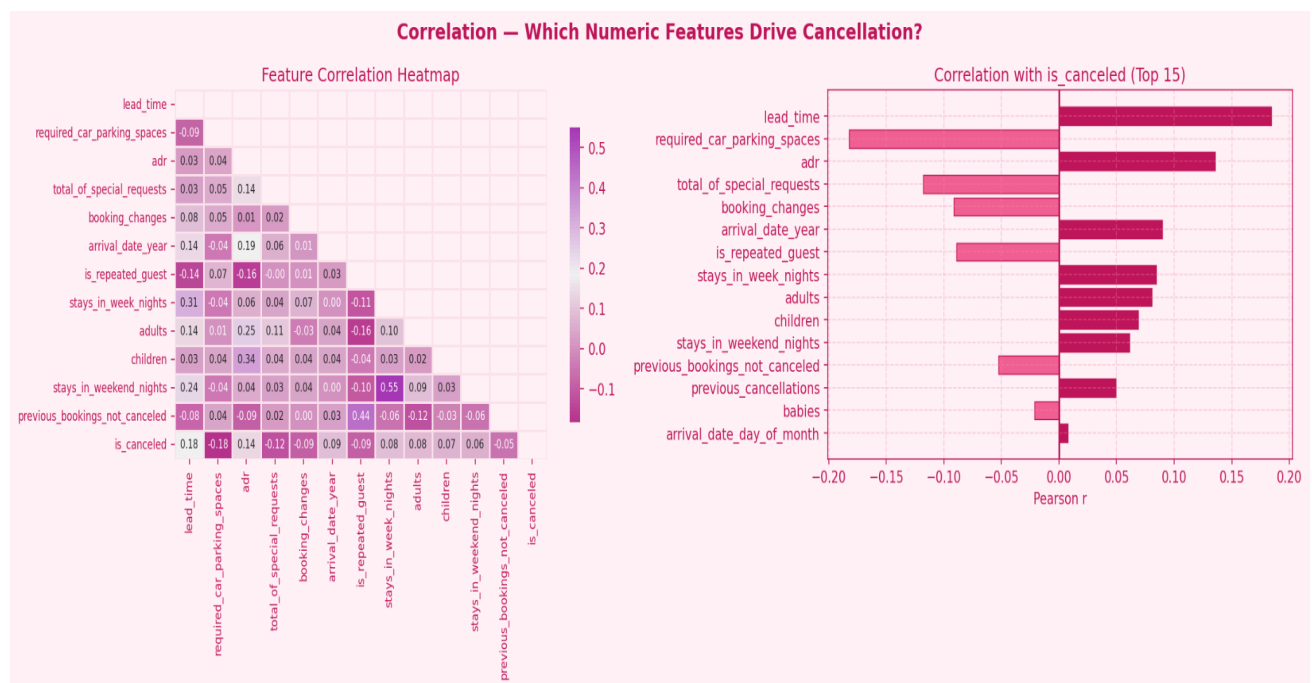


Figure 2 “ Correlation Analysis”

Lead Time and Cancellation

Cancellation rate increases steadily with lead time, rising from about **9% for bookings within a week** to around **41% for bookings made more than a year in advance**. This reflects greater uncertainty over longer planning horizons; the further in advance a booking is made, the higher the likelihood that plans change.

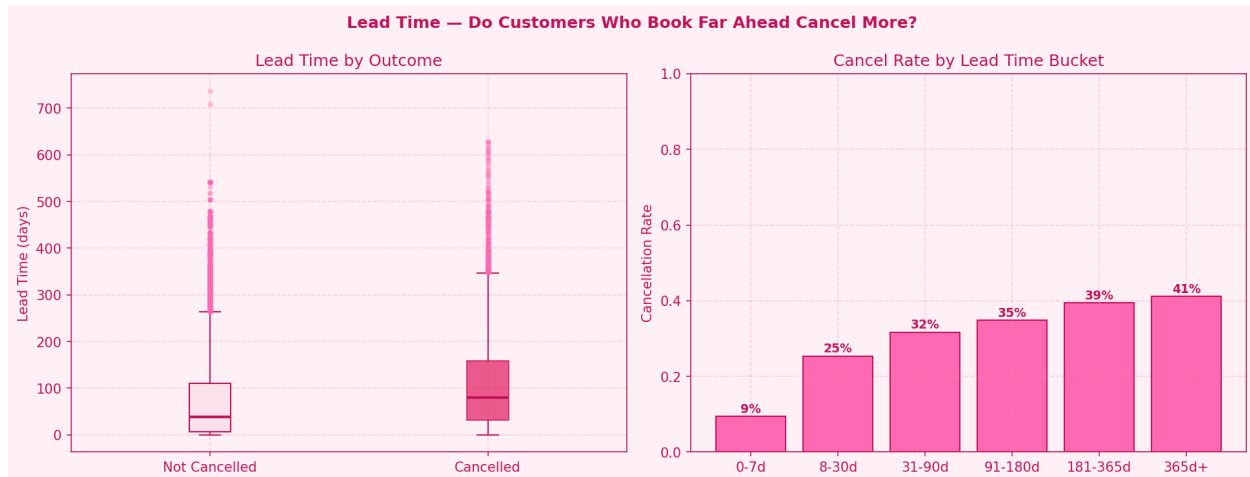


Figure 3” Lead Time vs Cancellation “

Business Recommendation: Bookings made more than 90 days in advance cancel at a rate exceeding 35%, rising to approximately 41% beyond one year. Hotels should implement a mandatory deposit policy for all reservations with a lead time greater than 90 days to financially secure these high-risk bookings at the time of reservation.

Price (ADR) and Cancellation

Cancellation risk generally increases with price. Bookings under €50 per night have a cancellation rate of about **15%**, rising to around **38% in the €200-300 range**, before slightly decreasing for the highest price tier. This suggests that higher-priced bookings are more likely to be cancelled, possibly due to greater flexibility or continued comparison with alternative options.

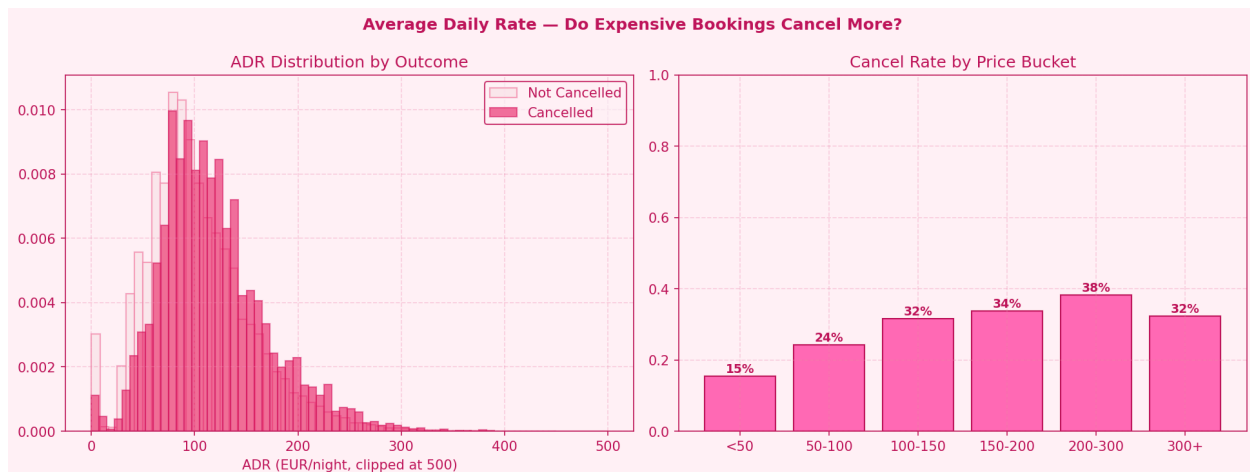


Figure 4 “ ADR and Cancellation “

Business Recommendation: Bookings priced above €200 per night cancel at rates between 32–38%, likely because customers continue comparing alternatives after reserving. Hotels could reduce this risk by offering small incentives — such as complimentary upgrades or breakfast inclusion — to guests who confirm expensive bookings, increasing their commitment to the stay.

Categorical Drivers of Cancellation

Deposit type is a strong signal. **Non-Refund deposits show an extremely high cancellation rate (around 95%)**, though this is based on a small number of bookings and should be interpreted with caution. In comparison, **No Deposit bookings cancel at around 28%**, while **Refundable bookings are slightly lower at about 25%**.

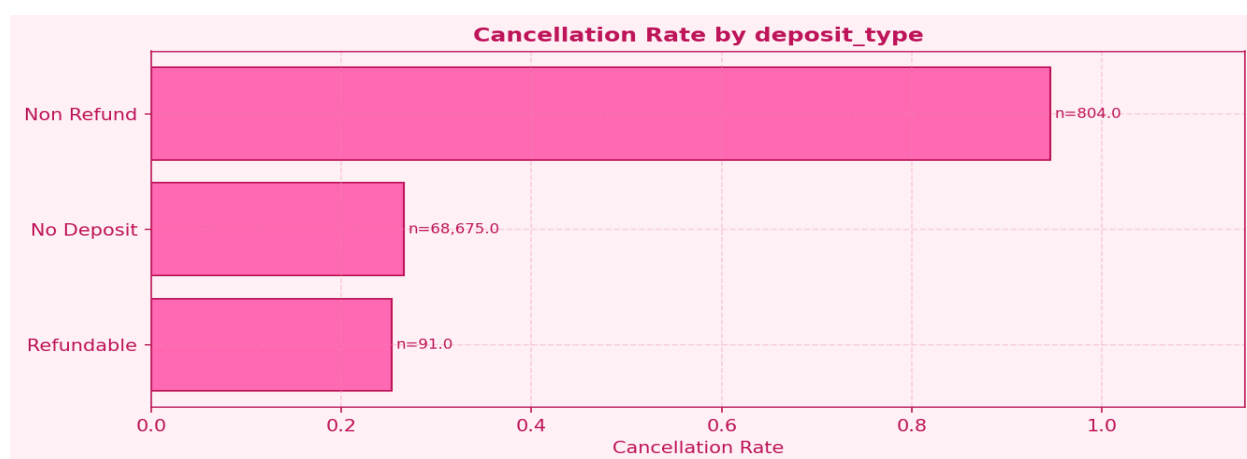


Figure 5a “Cancellation Rate by Deposit Type”

Business Recommendation: Non-Refundable deposit bookings show a near-100% cancellation rate, which reflects that this policy is selectively applied to already high-risk reservations rather than causing cancellations. Hotels should consider expanding the use of partial deposits ; not necessarily full non-refundable amounts , to bookings arriving through Online TA and Group channels, where cancellation risk is highest.

Cancellation rates vary significantly by market segment. **Online Travel Agencies (Online TA)** show the highest meaningful cancellation rate (around **35%**), followed by **Groups (~25%)** and **Aviation (~20%)**. In contrast, **Direct** and **Corporate** channels have much lower cancellation rates (around **15% or less**), confirming that booking channels are a strong indicator of cancellation risk. The Undefined segment (n=1) cancels at 100% but is statistically irrelevant.

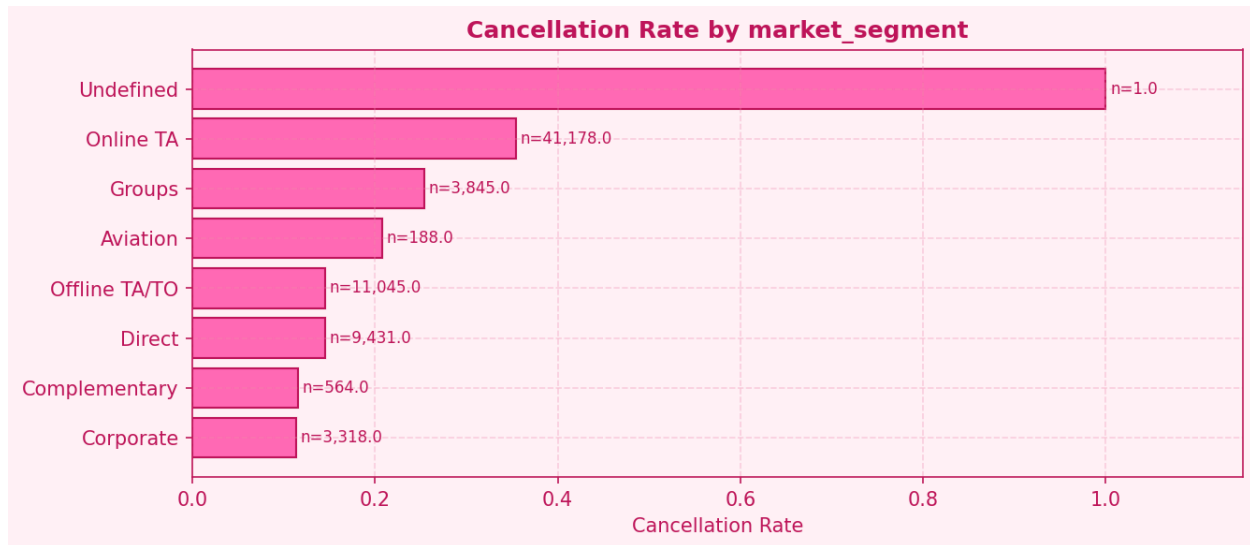


Figure 5b “ Cancellation Rate by Market Segment “

Business Recommendation: Direct and Corporate channels cancel at under 15% while Online TA exceeds 36%. Revenue managers should apply stricter cancellation policies and deposit requirements specifically to Online TA and Group reservations, while rewarding Direct and Corporate bookers with more flexible terms to encourage loyalty.

Transient customers have the highest cancellation rate at approximately 30% (n=57,385), while Group customers cancel the least at around 9% (n=439). Contract customers cancel at about 17% and Transient-Party at about 14%

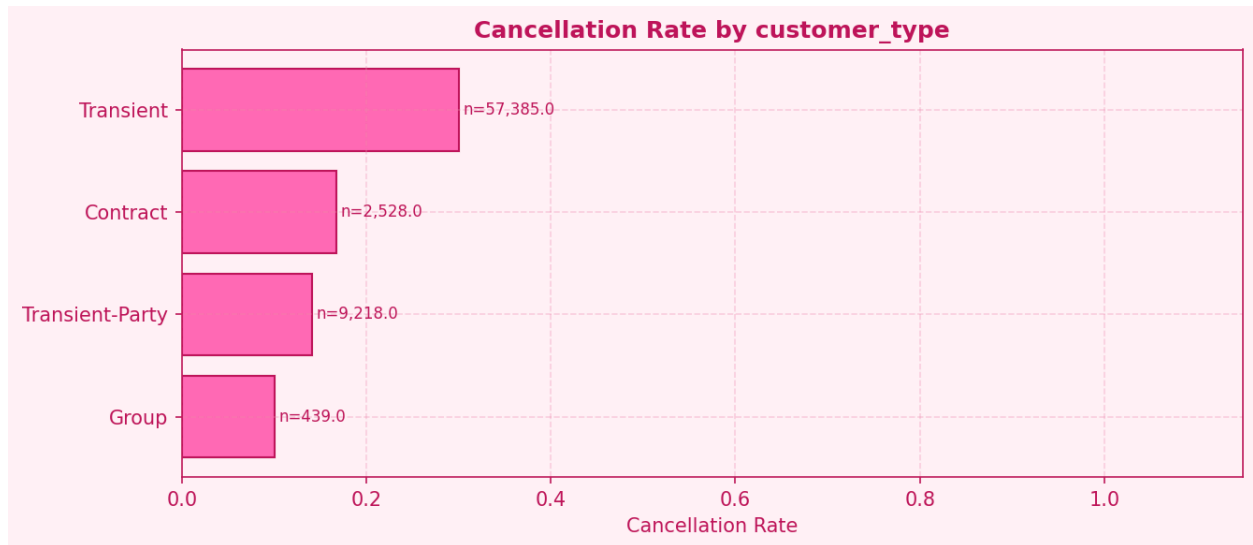


Figure 5c “ Cancellation Rate by Customer Type”

Business Recommendation: Transient customers cancel at approximately 30%, the highest rate among all customer types, and represent by far the largest segment with 57,385 bookings. Targeting this segment with personalized retention actions, such as pre-arrival engagement emails or early check-in incentives, could meaningfully reduce overall cancellation volume.

Seasonality

Booking volume peaks in summer, with **August reaching around 9,000 bookings**, while winter months are noticeably lower. Cancellation rates remain relatively stable throughout the year, ranging roughly from **20% to 30%**, with a slight increase during **spring and summer (April-August)**. This suggests that seasonality affects volume more than cancellation behavior, though peak periods still carry slightly higher risk.

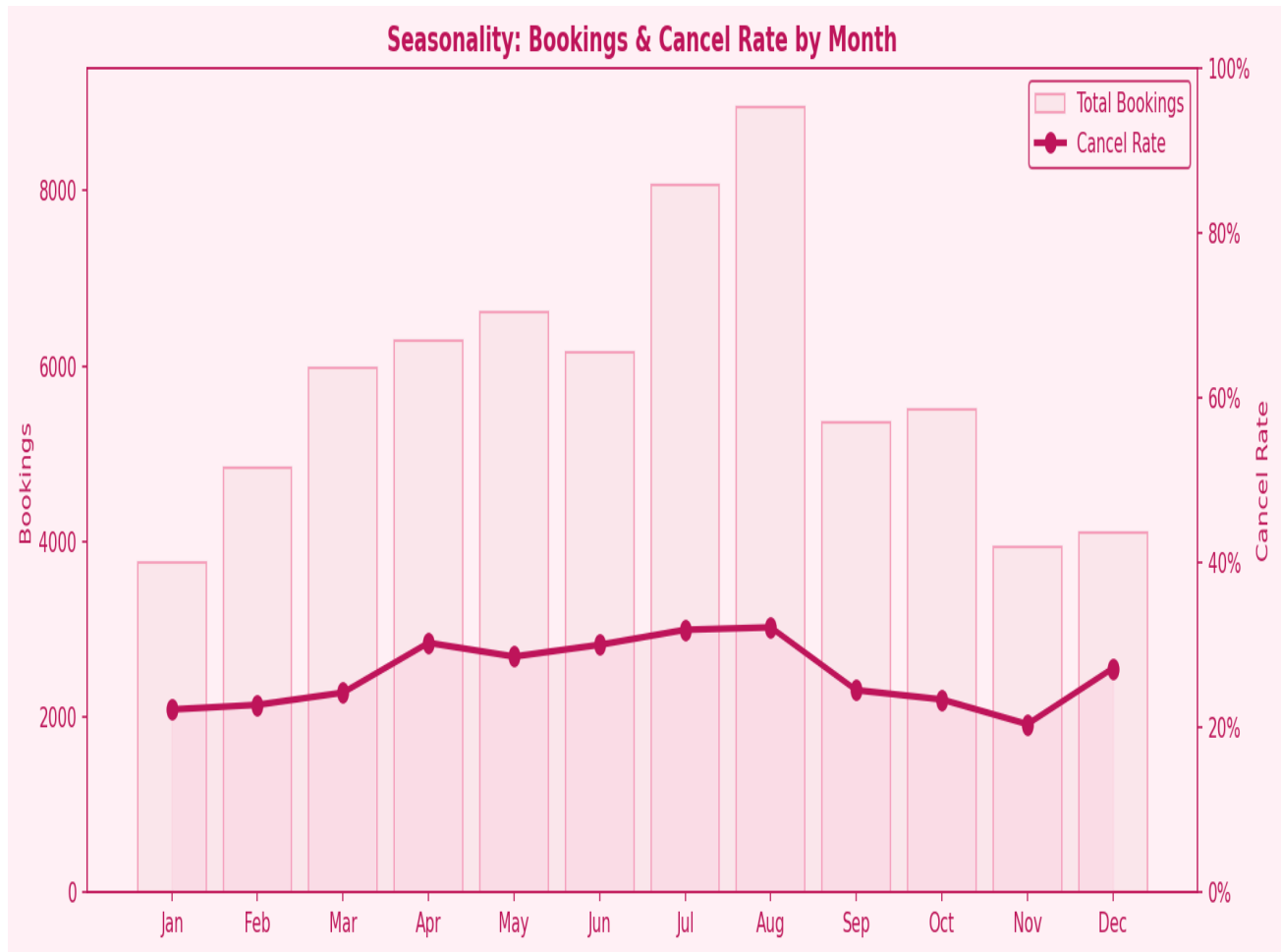


Figure 6” Seasonality: Bookings and Cancel Rate by Month “

Business Recommendation: Although booking volume peaks in August, cancellation rates remain elevated throughout the year, with April and May showing slightly higher rates than other months. Revenue managers should set larger overbooking buffers during April, May, and the summer months, and recalibrate these buffers dynamically using the model's risk scores rather than applying a single fixed overbooking rate year-round.

Skewness and Outliers

Several numeric features are strongly right-skewed, particularly `previous_cancellations` (skew=+34.2), `babies` (skew=+23.0), `adults` (skew=+22.3), `previous_bookings_not_canceled` (skew=+20.2), and `days_in_waiting_list` (skew=+19.7), with visible extreme outliers.

These distributions justify scaling for linear models like Logistic Regression, while tree-based models remain largely unaffected by skewness.

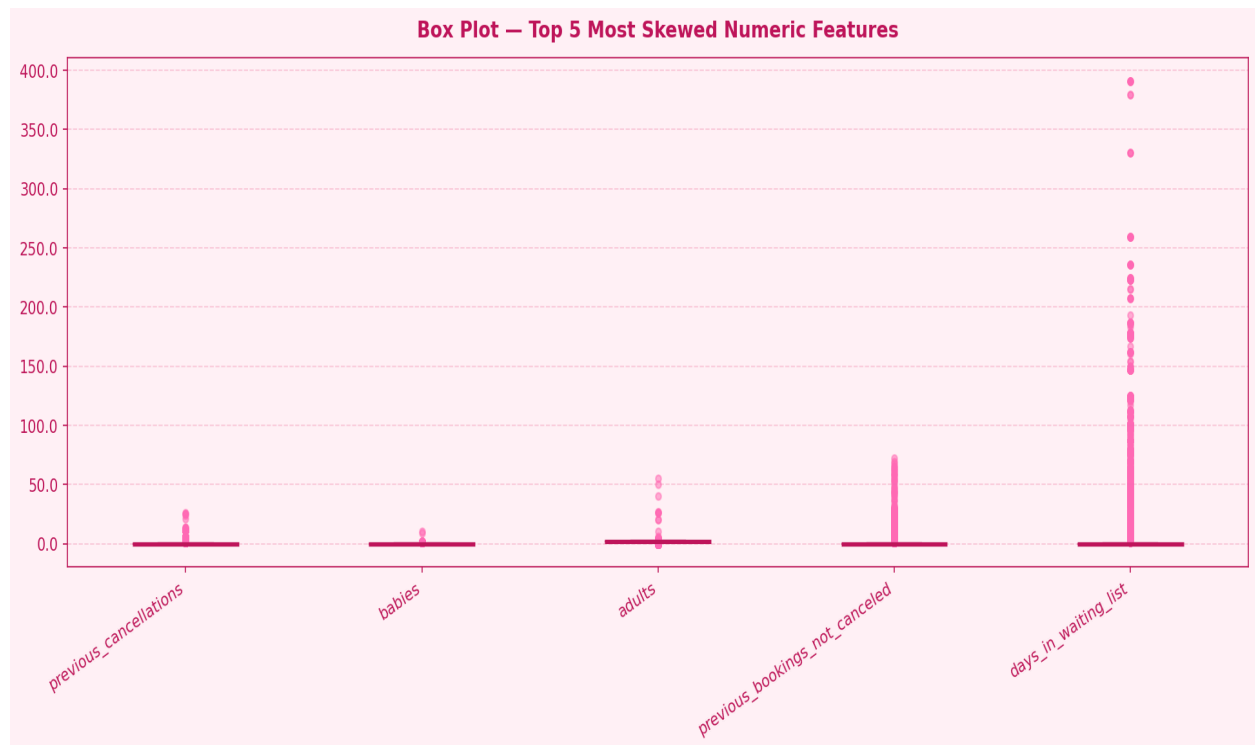


Figure 7 “Top 5 Most Skewed Features”

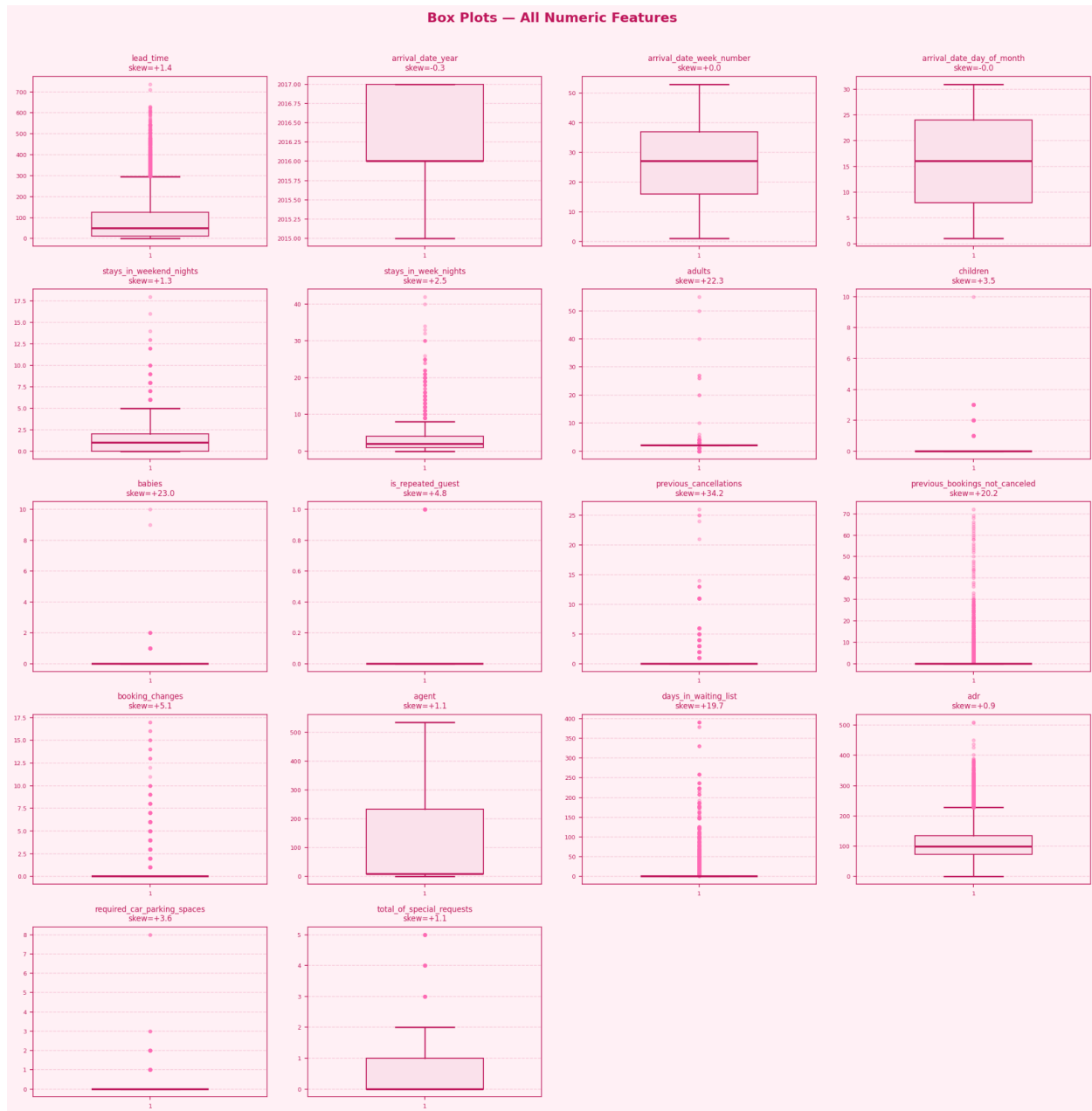


Figure 8 “Box Plots: All Numeric Features”

Methodological Note: Variables such as `previous_cancellations`, `days_in_waiting_list`, and `adults` contain extreme right-skewed distributions with significant outliers. `StandardScaler` was applied to normalise these for Logistic Regression. Random Forest is naturally robust to skewed distributions and outliers, so no transformation was required for tree-based models.

5.2 PCA Analysis

PC1 captures approximately **98.2%** of the total variance, while **PC2** contributes only **0.7%**, indicating that the vast majority of information in the transformed feature space is concentrated in a single dominant direction. The **95%** variance threshold is already reached at **PC1** alone, as confirmed by the cumulative explained variance plot. The 2D projection of PC1 versus PC2 shows substantial overlap between cancelled and non-cancelled bookings across the entire scatter space, meaning the two classes are not linearly separable in the reduced PCA representation. This confirms that dimensionality reduction via PCA does not improve class separability and explains why it was retained as a diagnostic tool rather than adopted as a preprocessing step for the final models.

The loadings heatmap reveals that PC1 is almost entirely driven by a single encoded variable (`ord_country`, loading ≈ 1.00), which explains why it dominates variance without providing useful discriminative signal. The more informative structure emerges in PC4 and PC5: PC4 is shaped by combinations of `num_lead_time` (0.38), `num_stays_in_week_nights` (0.42), `num_previous_bookings_not_canceled` (-0.38), `num_stays_in_weekend_nights` (0.40), and `num_is_repeated_guest` (-0.41), while PC5 is driven by `num_previous_cancellations` (0.38), `num_lead_time` (0.26), `num_stays_in_week_nights` (0.44), `num_previous_bookings_not_canceled` (0.43), `num_stays_in_weekend_nights` (0.42), and `num_is_repeated_guest` (0.33). These later components link the PCA structure back to behaviorally meaningful features: stay duration, booking history, and guest loyalty, which are also among the top predictors identified by Random Forest.

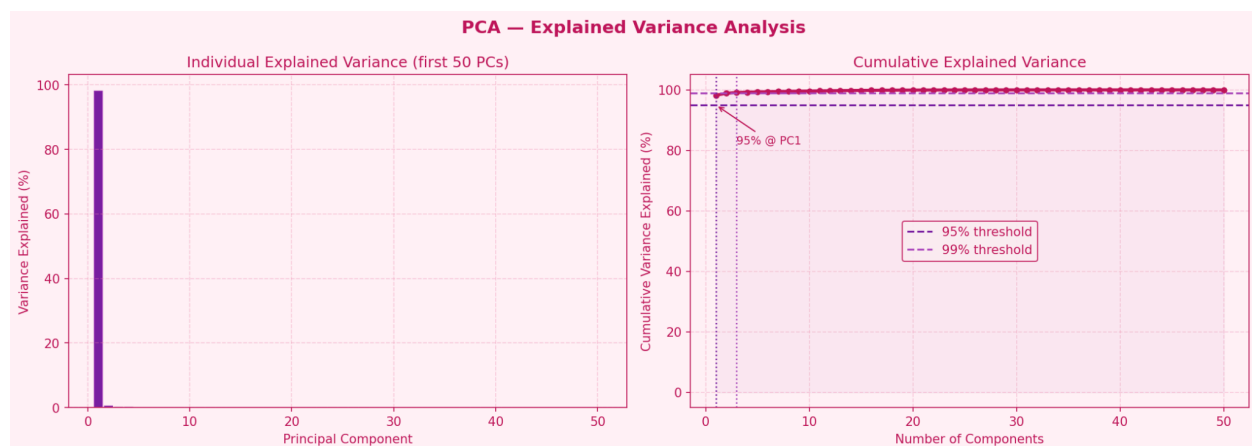


Figure 9” PCA Explained Variance “

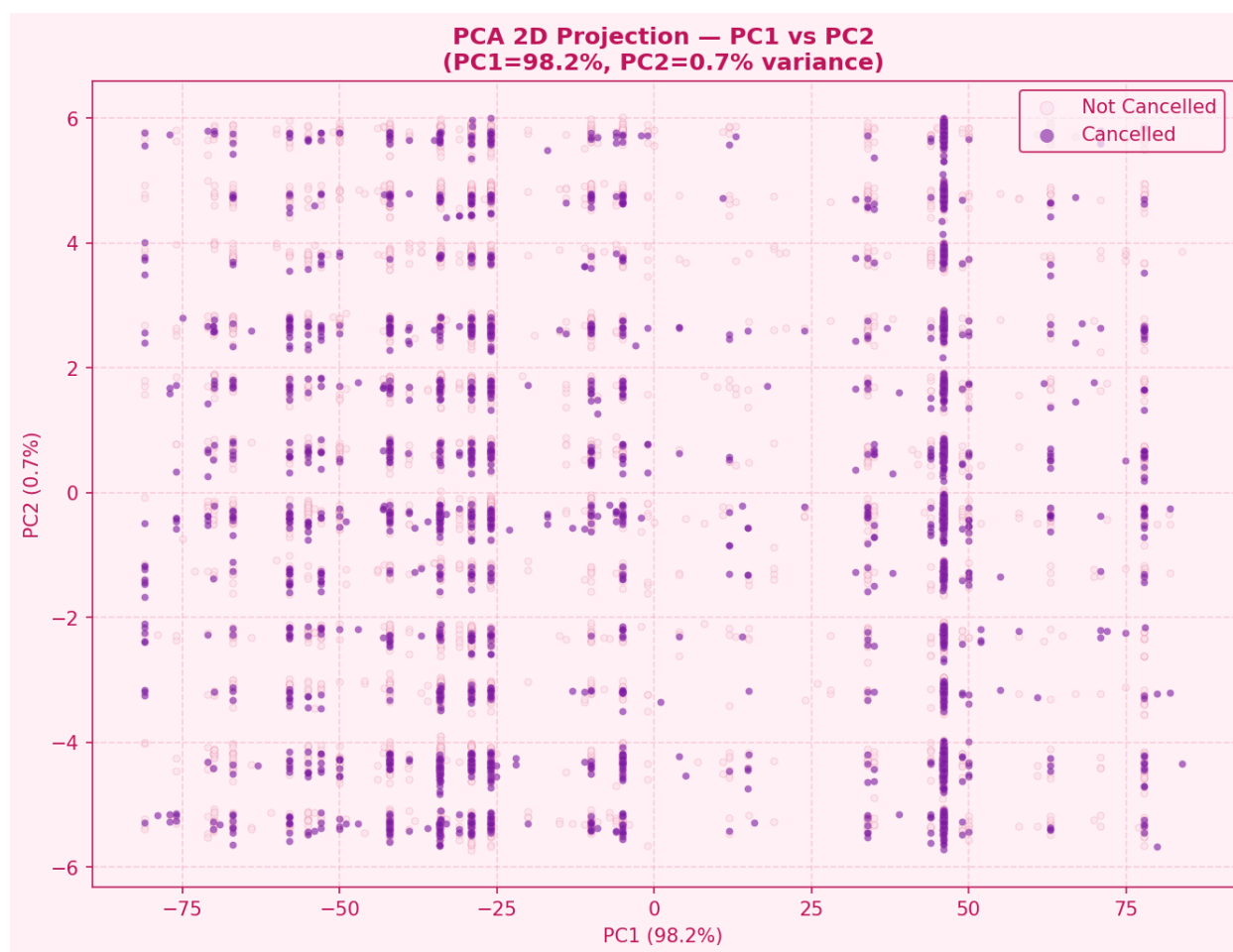


Figure 10 “PCA 2D Projection”

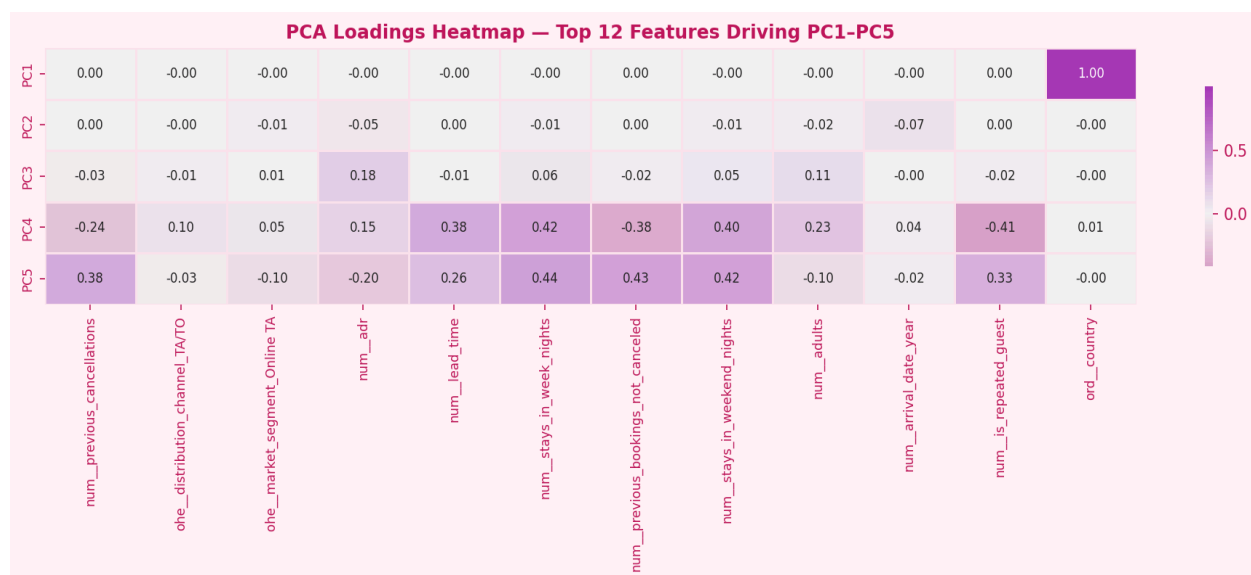


Figure 11” PCA Loadings Heatmap”

5.3 Model Comparison

Random Forest delivers the best overall performance, achieving a Test AUC of 0.892 and the highest F1-score (0.649), indicating strong ability to identify cancellations while maintaining a good balance between precision and recall. Decision Tree follows with a Test AUC of 0.866 and an F1-score of 0.626, while Logistic Regression provides a lower but stable baseline (Test AUC = 0.814, F1 = 0.517).

The close alignment between CV AUC and Test AUC across all models suggests that the models generalize well and are not overfitting. The superior F1-score of Random Forest confirms that it handles the class imbalance more effectively, making it the most suitable model for this problem.

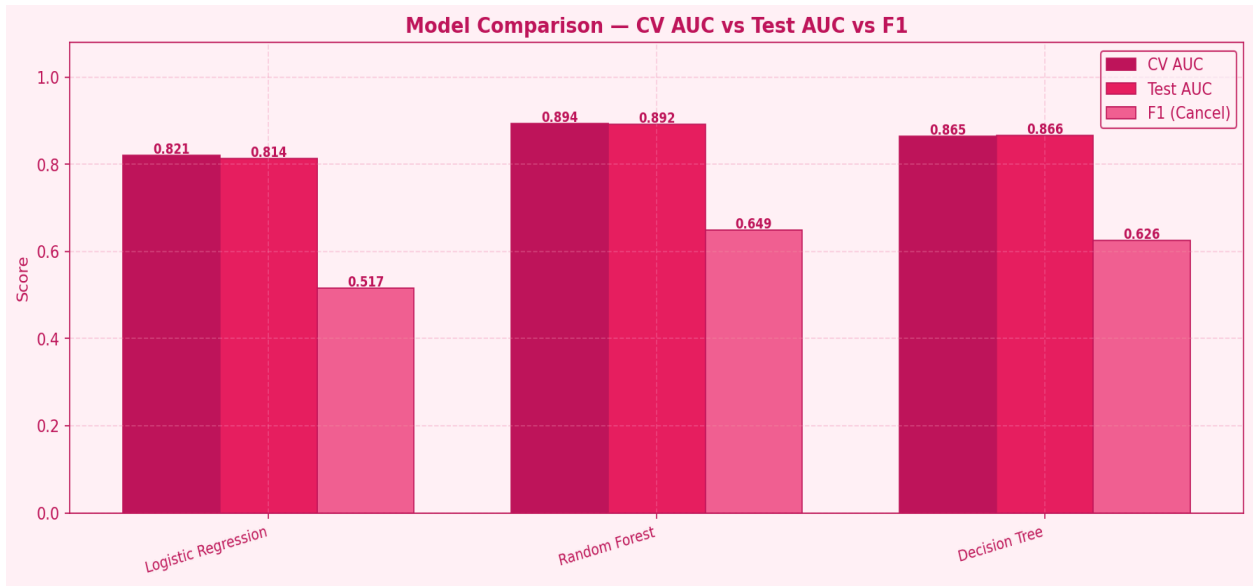


Figure 12 “ Model Comparison “

Model	CV AUC	Test AUC	F1 (Cancelled)
Logistic Regression	0.821	0.814	0.517
Random Forest	0.894	0.892	0.649
Decision Tree	0.865	0.866	0.626

5.4 Error Types and Business Trade-off

In this classification problem, two types of errors are critical. A false positive (Type I error) occurs when a booking is predicted as cancelled but is actually honoured, potentially leading to unnecessary overbooking or lost customer trust. A false negative (Type II error) occurs when a cancellation is not predicted, resulting in an empty room and direct revenue loss.

The results show that Random Forest achieves the lowest false positive rate (6.6%), outperforming both Logistic Regression (8.1%) and Decision Tree (10.4%). However, false negative rates remain substantially higher across all models, particularly for Logistic Regression (57.7%), compared to Random Forest (43.6%) and Decision Tree (41.9%).

Given that missed cancellations (false negatives) are more costly in practice, this imbalance highlights the need to prioritise recall for the cancellation class. This trade-off is further addressed through threshold tuning, allowing the model to better capture high-risk bookings.

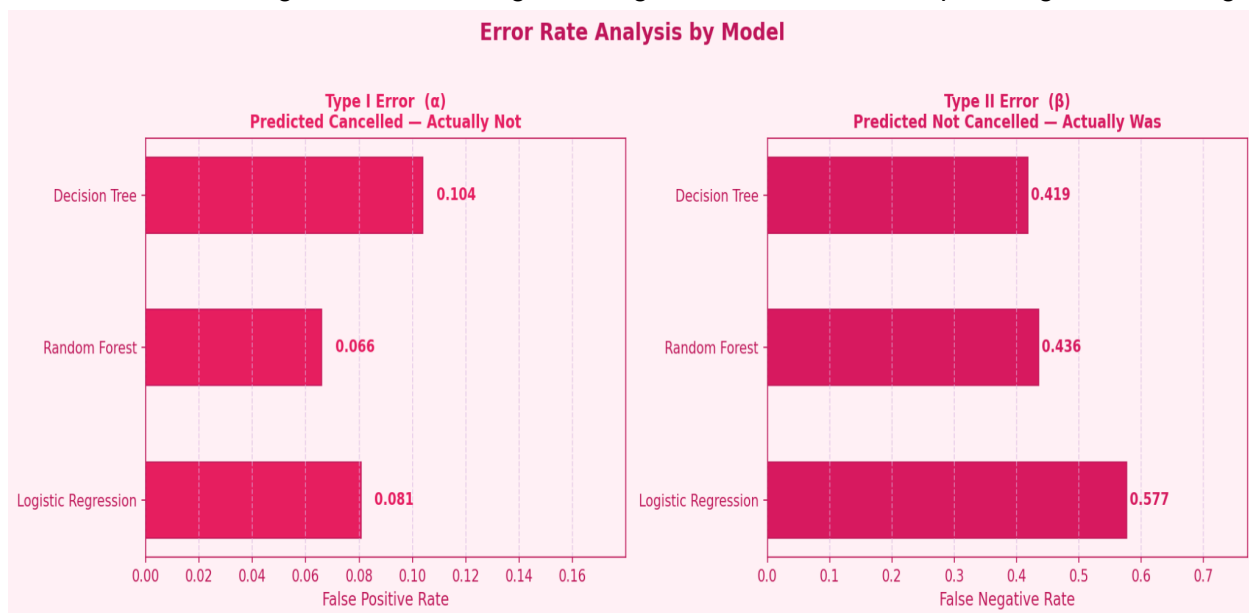


Figure 13 “ Error Rate Analysis by Model”

5.5 ROC Curves

The ROC curves show that Random Forest achieves the best overall performance with an AUC of 0.89, consistently dominating across all classification thresholds. Decision Tree follows closely with an AUC of 0.87, while Logistic Regression performs noticeably worse with an AUC of 0.81.

The gap between the curves is consistent rather than marginal, confirming that non-linear models capture the structure of cancellation behaviour more effectively than linear models. In particular, Random Forest maintains higher true positive rates at lower false positive rates, making it the most reliable model for identifying cancellations.

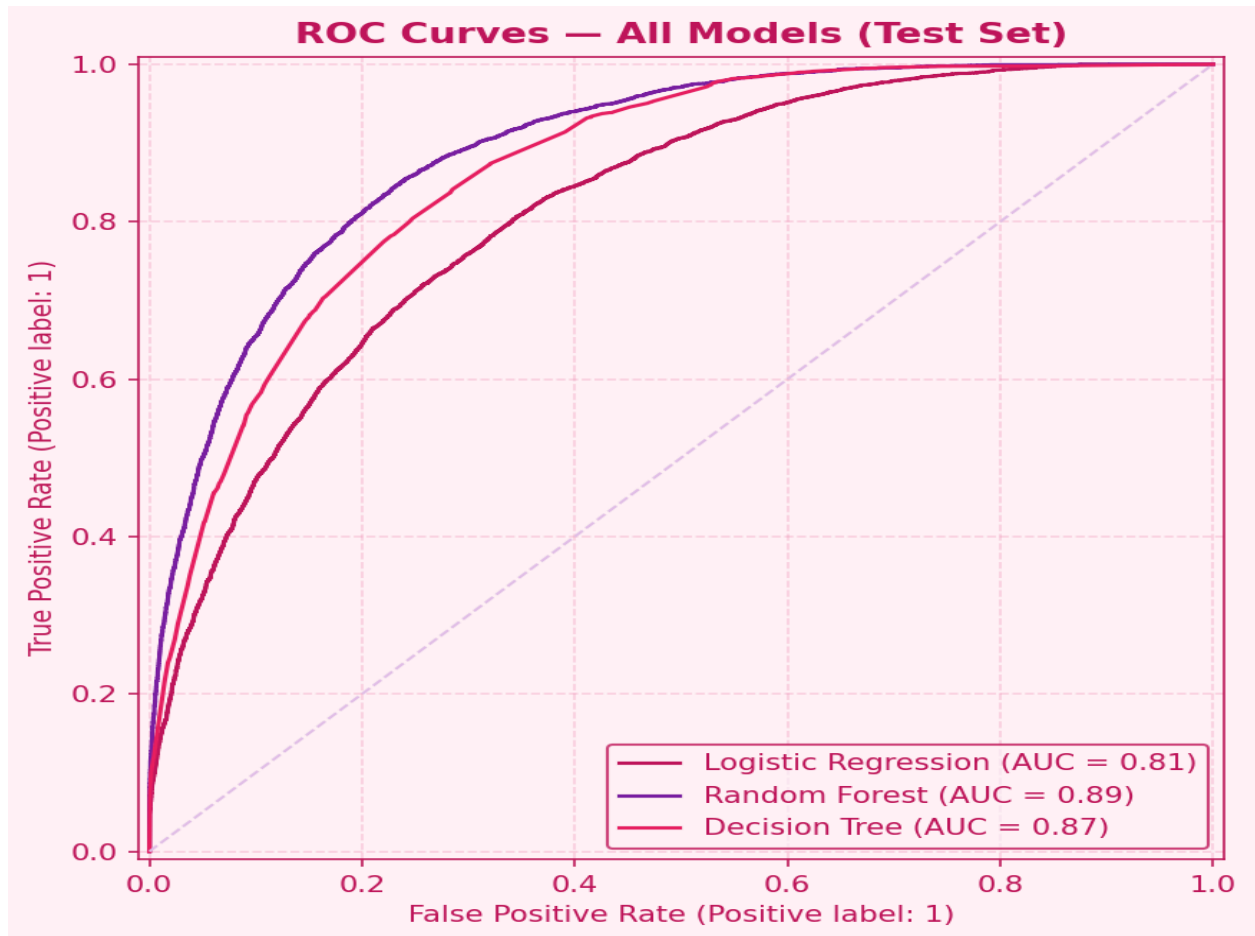


Figure 14 "ROC Curves"

5.6 Confusion Matrix

The Random Forest model correctly identified 11,808 non-cancelled bookings and 2,681 cancellations on the test set. It missed 2,069 cancellations (false negatives) and produced 835 false positives. False negatives are the more costly error in practice, as each one represents a cancellation the hotel failed to anticipate. This trade-off can be adjusted by tuning the decision threshold, as demonstrated in **Section 5.8**.

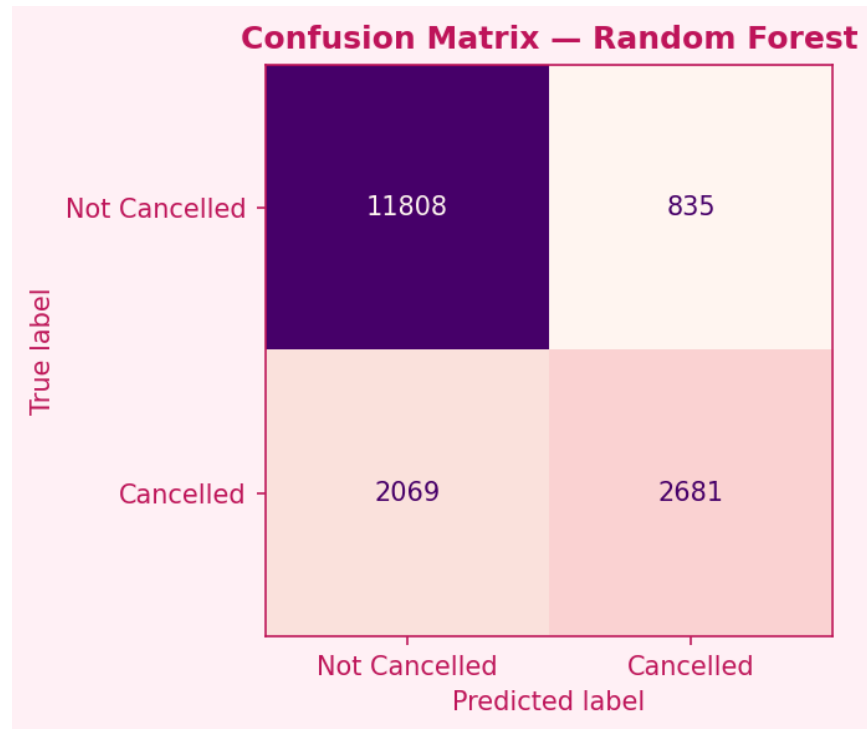


Figure 15 “ Confusion Matrix (Random Forest)”

5.7 Feature Importance

The most influential features are `num_lead_time`, `ord_country`, `num_total_of_special_requests`, and `num_adr`, followed by `ohe_market_segment_Online TA`, `num_required_car_parking_spaces`, `num_stays_in_week_nights`, `ord_arrival_date_month`, `num_arrival_date_year`, `num_previous_cancellations`, `ohe_market_segment_Offline TA/TO`, `ohe_customer_type_Transient`, `ord_assigned_room_type`, `num_booking_changes`, and `ohe_deposit_type_Non Refund`. This closely aligns with the EDA findings, confirming that the model is capturing meaningful behavioural patterns rather than statistical noise. In particular, `total_of_special_requests` stands out as an important negative risk signal because customers who make special requests are more engaged and therefore less likely to cancel. Similarly, longer `lead_time` and higher `adr` are associated with increased cancellation risk, reinforcing earlier insights.

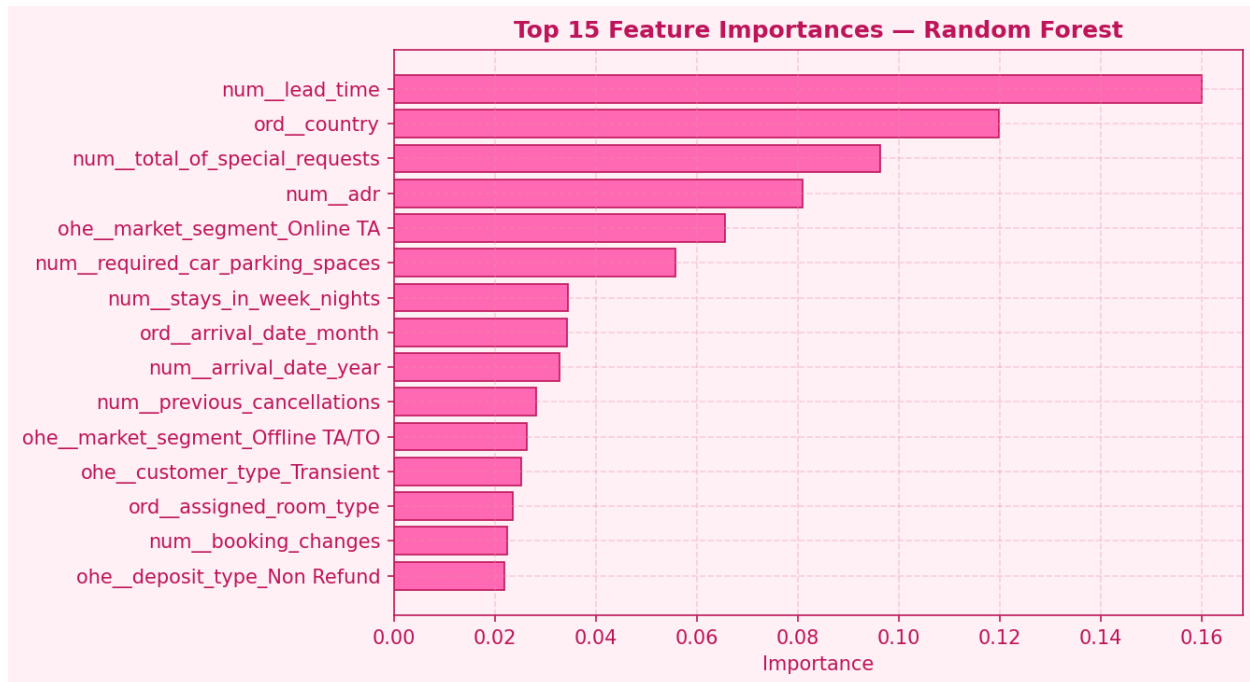


Figure 16 “ Top 15 Feature Importances (Random Forest) “

Business Insight: Lead time is the dominant predictor by a significant margin, confirming the EDA finding. The high importance of total_of_special_requests is particularly noteworthy, guests who make special requests are more invested in their stay and significantly less likely to cancel. Hotels could actively encourage guests to add special requests at the time of booking as a low-cost strategy to increase commitment and reduce cancellation risk. The presence of country and agent as top predictors also suggests that cancellation risk varies meaningfully by guest origin and booking channel, which should inform targeted policy decisions.

5.8 Specificity, Sensitivity and Threshold Tuning

5.8.1 Specificity and Sensitivity

Beyond AUC, each model was evaluated on two operationally critical metrics: Sensitivity (Recall the proportion of actual cancellations correctly identified) and Specificity (the proportion of actual non-cancellations correctly identified). These two metrics map directly to the two types of business error: low Sensitivity means the hotel misses real cancellations and loses revenue; low Specificity means the hotel raises unnecessary alarms and contacts guests who were never going to cancel.

Model	Sensitivity	Specificity	Type I Error	Type II Error
Logistic Regression	42.3%	91.9%	8.1%	57.7%
Random Forest	56.4%	93.4%	6.6%	43.6%
Decision Tree	58.1%	89.6%	10.4%	41.9%

Random Forest achieves the highest Specificity (93.4%) and the lowest Type I Error (6.6%), making it the most conservative model in terms of false alarms. Decision Tree achieves slightly higher Sensitivity but at the cost of notably worse Specificity and a higher Type I Error rate. This confirms Random Forest as the most balanced and operationally reliable model overall.

5.8.2 Threshold Tuning

The default decision threshold of 0.50 was designed for balanced classes. Since cancellations represent only 27% of bookings, this threshold is biased toward predicting "Not Cancelled," causing the model to systematically underestimate genuine cancellation risk. To correct this, the Random Forest's predicted probabilities were swept across thresholds from 0.10 to 0.90, tracking F1-score, Recall, Type I Error, and Type II Error at each point. The optimal threshold was identified at $t = 0.31$, at which point F1 reaches 0.694 and Recall reaches 0.801. At this threshold, the model correctly identifies 3,806 cancellations compared to only 2,681 at the default threshold, while false negatives drop from 2,069 to 944, a substantial operational improvement.

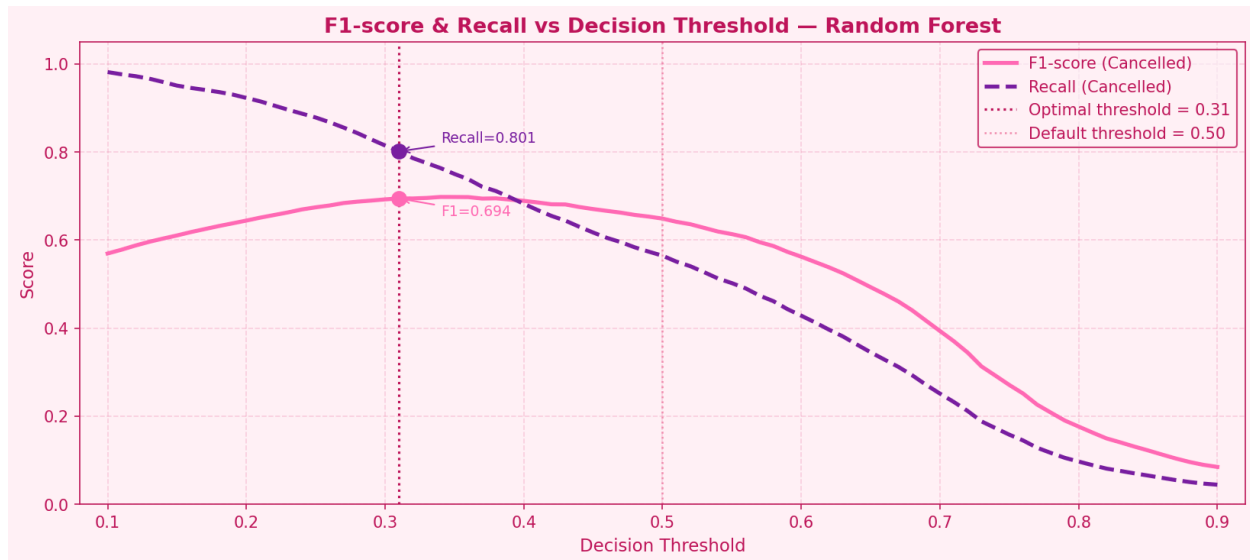


Figure 13a "F1-score & Recall vs decision threshold"

Business Recommendation: With 80% of cancellations now detected, revenue management can trigger early interventions like flexible rebooking offers or overbooking buffers , before rooms go unsold. The F1 plateau around $t = 0.31$ also signals that pushing recall higher would cost more in false alarms than it saves.

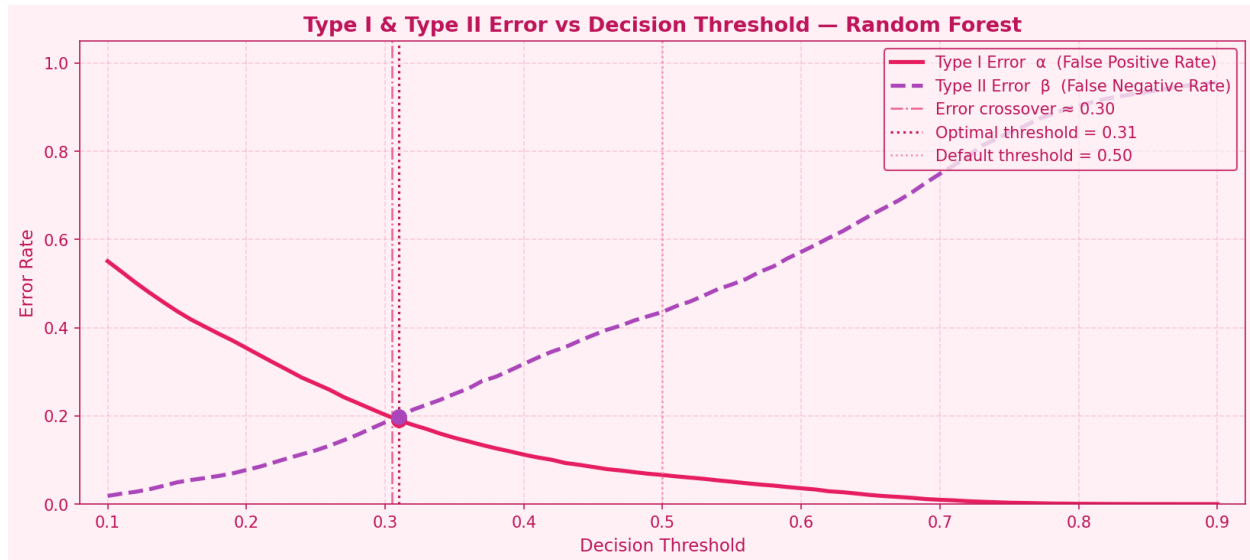


Figure 13b "Type I and Type II error vs decision threshold"

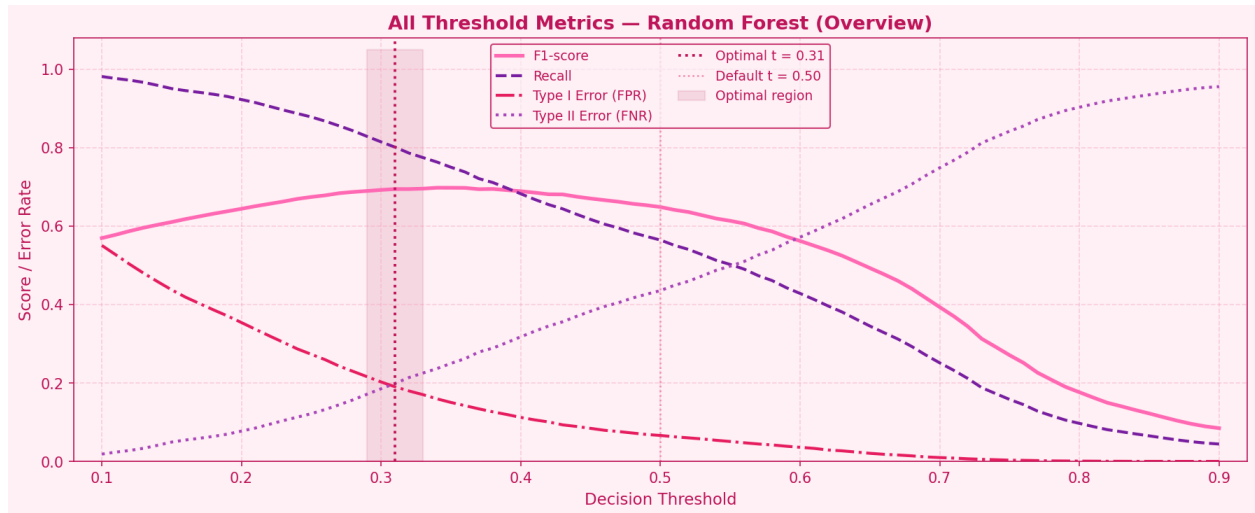


Figure 13c "All threshold metrics"

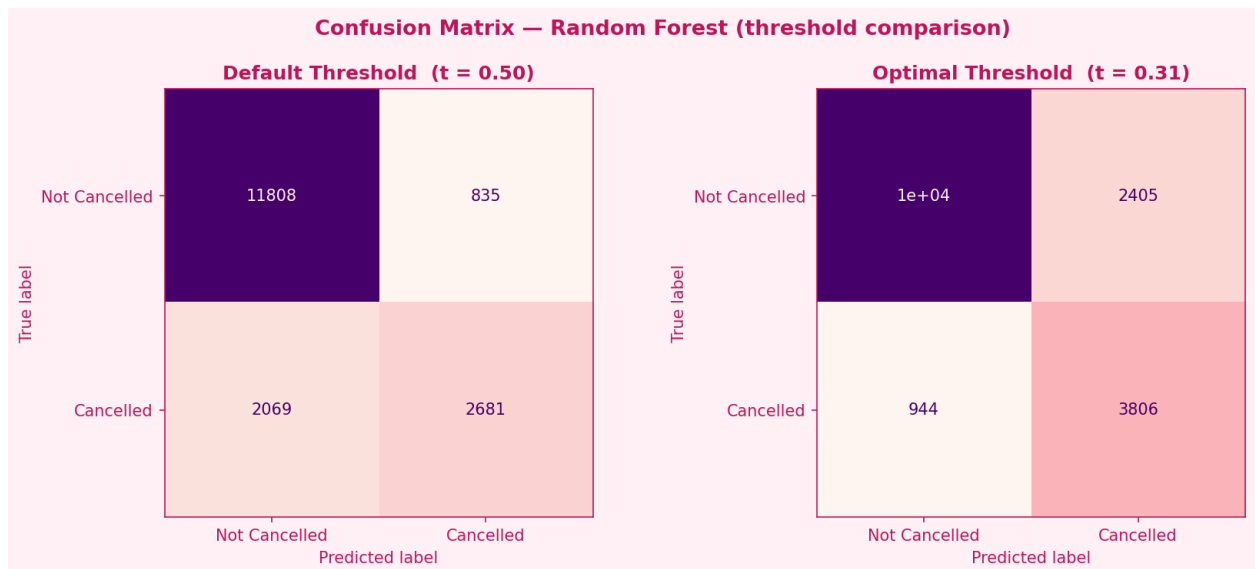


Figure 13d "Confusion matrix ,threshold comparison"

Business Interpretation: In hotel revenue management, a missed cancellation (Type II error) is more costly than a false alarm (Type I error). A missed cancellation means a room goes unsold with no opportunity to intervene; a false alarm means an unnecessary action is taken for a guest who would have stayed anyway at a much smaller operational cost. Lowering the threshold from 0.50 to 0.31 reduces missed cancellations from 2,069 to 944 at the cost of increasing false alarms from 835 to 2,405. This is a trade-off that is clearly justified when the revenue impact of an empty room far exceeds the cost of a precautionary outreach to a guest who ultimately stays.

6. Conclusion

This project set out to predict hotel booking cancellations and identify their main drivers. A rigorous pipeline was implemented, from leakage prevention and stratified preprocessing to cross-validated model comparison, ensuring that the results reflect a genuine predictive signal rather than methodological artifacts.

Three key findings emerge. First, `lead_time` is the dominant predictor: bookings made more than 90 days in advance exceed a 35% cancellation rate, making deposit requirements for long lead-time reservations the single most actionable policy recommendation. Second, contextual variables such as `deposit_type`, `market_segment`, and `customer_type` provide strong complementary signals: Online TA channels cancel at over 36%, while direct and corporate channels stay under 15%; transient customers represent both the highest cancellation rate and the largest booking segment, making them the primary target for retention actions. Third, Random Forest outperforms all other models (Test AUC = 0.89), confirming that cancellation behavior is driven by non-linear interactions that linear models cannot fully capture. Threshold tuning further improves the model's operational value: lowering the decision boundary from 0.50 to 0.31 reduces missed cancellations from 2,069 to 944, a trade-off explicitly justified by the asymmetric cost of missed cancellations in hotel revenue management.

From a business perspective, hotels should consider three concrete interventions: require deposits for all bookings with lead times exceeding 90 days; apply stricter cancellation policies to online TA and group segments while rewarding direct and corporate bookers with flexible terms; and actively encourage guests to submit special requests at booking time, since this signals commitment and is strongly associated with lower cancellation risk. These recommendations are not introduced here for the first time; they emerge directly from the EDA findings documented in **Section 5.1** and are confirmed by the feature importance results in **Section 5.7**.

7. Who Benefits from This Model

This work primarily benefits **hotels**, where revenue managers, operations teams, and marketing departments can use cancellation predictions to optimize pricing, overbooking, and customer engagement strategies. It also provides value to **online travel agencies** such as **Booking.com** and **Expedia**, which can improve recommendation systems and better manage booking risks across listings. **Large hotel chains** like **Marriott International** and **Hilton** benefit at scale through more accurate demand forecasting and global pricing strategies. Additionally, **consulting firms** such as **EY** can leverage this as a real-world predictive analytics case, while **financial & insurance institutions** can use similar models to assess revenue risk and design products related to cancellation protection.

8. Limitations

The dataset covers only two hotels in Portugal, which limits how broadly the findings generalize. The relationships identified are associative, not causal. Feature selection relied on linear correlation, which may have discarded features with non-linear predictive value. Finally, the models were evaluated on historical data and were not tested in a live deployment setting where booking patterns shift over time.

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